



On the Learnability of Test-Time Adaptation

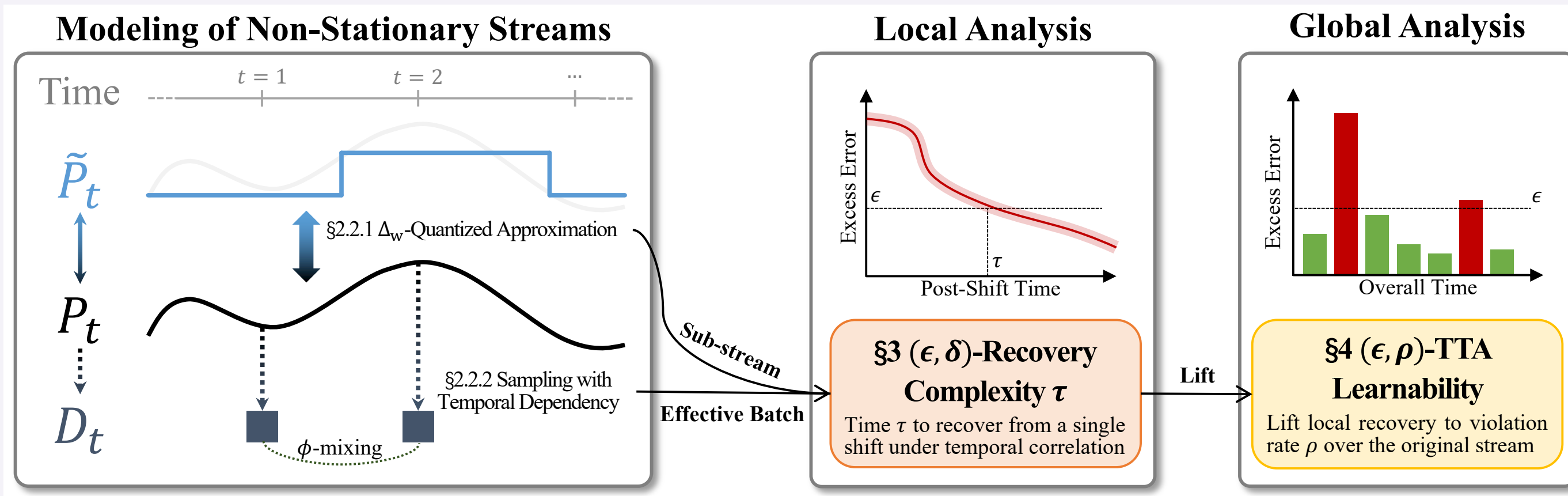
A Recovery Complexity Perspective

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TL;DR: The first learnability theory for TTA on non-stationary streams — defining (ϵ, δ) -Recovery Complexity (with matching bounds) and (ϵ, ρ) -TTA Learnability.

Framework at a Glance



Unified stream model → **local** recovery → **global** learnability → regret.

Motivation & The Gap

Test-time adaptation (TTA) adapts a source model to distribution shifts using **only unlabeled** data via a *proxy loss* ψ — empirically successful, yet it can **collapse** under complex shifts.

Why existing theory falls short:

- **Online learning** controls *average* (regret) performance — not the **instantaneous, per-step reliability** TTA demands.
- Prior TTA bounds assume **special algorithms or label access** — no *global* guarantee for unlabeled supervision, non-stationarity, and correlation.

This work: a principled framework answering *when is TTA learnable?*

Modeling the Non-Stationary Stream

A stream $S = \{D_t\}_{t=1}^T$, $D_t \sim \mathcal{P}_t$, is captured along **two** axes.

(1) **Global distribution shift** — a W_1 -quantized surrogate turns a continuous trajectory into a **piecewise-constant** stream (one knob Δ_W unifies **gradual & abrupt** shifts):

$$V_T = \sum_t W_1(\mathcal{P}_t, \mathcal{P}_{t+1}), \quad \text{\#shifts } \tilde{K}_S(T) \leq \left\lceil \frac{2V_T}{\Delta_W} \right\rceil$$

(2) **Local temporal correlation** — a ϕ -mixing process ($\phi(i) \leq \varrho^i$) **shrinks the effective batch size** ($C_\phi=1$ iff independent):

$$B_{\text{eff}} = \frac{B}{C_\phi} \leq B, \quad C_\phi = 1 + \frac{4\sqrt{\varrho}}{1 - \sqrt{\varrho}}$$

A Well-Defined Target: Proxy-Optimal Competitor

Perfect adaptation is unattainable when $\psi \neq \ell$. We compete against the best **proxy-optimal** model in a local neighborhood $\mathcal{N}_r(\theta_1)$, matching the *constrained* adaptation of practical TTA methods:

$$\theta_t^* \in \arg \min_{\theta \in \mathcal{N}_r(\theta_1)} \psi_t(\theta), \quad R_t := \ell_t(\theta_t^*), \quad \mathcal{E}_t := \ell_t(\theta_t) - R_t$$

The **excess risk** $\mathcal{E}_t$ is the central quantity throughout. It is driven down through the proxy only under (α, ζ) -**alignment**:

$$\langle \nabla \psi_t, \nabla \ell_t \rangle \geq \alpha \|\nabla \ell_t\|^2 - \zeta$$

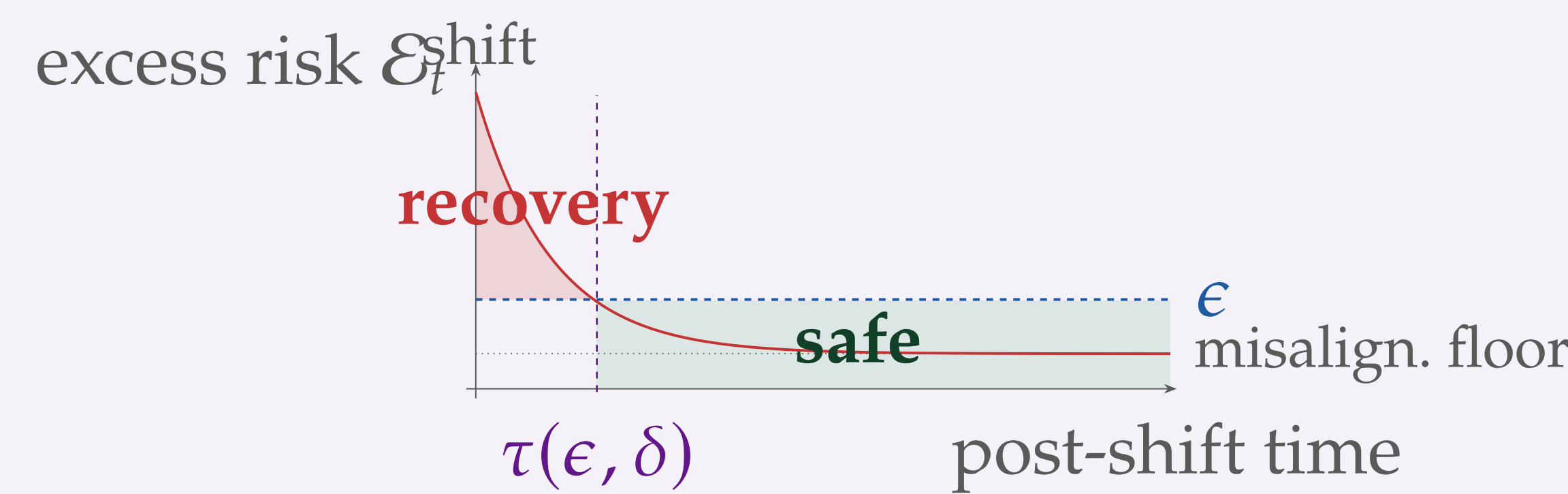
$\alpha > 0$ (alignment strength) gives a descent signal; $\zeta \geq 0$ (irreducible misalignment) sets an error floor that no amount of data can remove.

Local View: (ϵ, δ) -Recovery Complexity

The **post-shift time** to drive excess risk below ϵ (prob. $\geq 1 - \delta$) and keep it there:

$$\tau(\epsilon, \delta) := \inf \left\{ t : \sup_{u \geq t} \mathbb{P}(\mathcal{E}_u > \epsilon) \leq \delta \right\}$$

Unlike regret, it controls the **length of unsatisfactory periods**.



Matching bounds (order-wise tight):

Lower (any algorithm, info. limit)

$$\tau \geq \Omega\left(\frac{C_\phi}{B \alpha (\sqrt{\zeta} + 2\alpha\epsilon + \sqrt{\zeta})^2}\right)$$

Upper (simple SGD baseline)

$$\tau \leq O\left(\frac{C_\phi}{B \alpha^2 \epsilon} \log \frac{\Delta_W + \epsilon}{\epsilon}\right)$$

Match up to log $\Rightarrow$ **the baseline is near-minimax-optimal**.

What governs recovery:

- **Error floor:** $\zeta > 0$ leaves a non-vanishing term as $\epsilon \rightarrow 0$ — misalignment caps accuracy.
- **Speed:** $\tau \propto 1/\alpha^2$ (perfect proxy) and $\tau \propto C_\phi/B$; both B and temporal correlation matter.
- Lower & upper bounds share $C_\phi/(B\alpha^2\epsilon)$ — no algorithm can do much better.

Global View: (ϵ, ρ) -TTA Learnability

(ϵ, ρ) -**learnable** if some algorithm keeps the violation fraction $\leq \rho$:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{P}(\ell_t(\theta_t) - R_t > \epsilon) \leq \rho$$

Recovery $\Rightarrow$ Learnability transfer ($\Lambda=L_x+GL_{\nabla\psi}/\mu$):

$$\rho \leq \delta + \frac{(\tilde{K}_S(T) + 1) \tau(\epsilon', \delta)}{T}, \quad \epsilon' = \epsilon - \Lambda \Delta_W$$

Strong transfer: faster recovery / fewer shifts $\Rightarrow$ a smaller ρ .

Connection to dynamic regret: $\text{Reg}(T) \leq T(\epsilon + M\rho)$; persistent shifts ($\Delta_W=\Omega(1)$) force **linear regret**, while the benign regime recovers classical $o(T)$.

Empirical Validation

Synthetic (1-D recovery, vary α at $B=16$): τ tracks the predicted $O(1/\alpha^2)$ scaling, between the lower (LB) and upper (UB) bounds.

α	LB	τ	UB
0.05	59.0	322.0	311.9
0.10	15.6	77.0	78.0
0.20	4.1	19.0	19.5
0.50	0.7	4.0	3.1

$\tau \cdot \alpha^2 \approx \text{const} \Rightarrow$ **near-tight**. ✓

Real (ImageNet-C, temporal correlation $\beta \downarrow$): $\tilde{\alpha}$ decides whether Tent **improves** or **collapses**.

β	$\tilde{\alpha}$	$\Delta \text{Acc.}$	$\# \text{Improve}$
0.001	-0.028	-12.7%	0/15
0.01	+0.090	+3.6%	12/15
0.1	+0.157	+13.7%	15/15
Uniform	+0.171	+14.7%	15/15

Alignment α is the single factor governing whether TTA succeeds.

Takeaway: Recovery complexity is the building block of TTA learnability — pinning down *when* adaptation is possible and *what* limits it.



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